Data Engineer

Preethi Ambati

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PROFESSIONAL SUMMARY

- Dynamic and results-oriented Data Engineer with 5+ years of experience designing, building, and optimizing large-scale data solutions across AWS, Azure, and GCP ecosystems.
- Proven expertise in building and optimizing petabyte-scale data pipelines, re-engineering workflows for improved efficiency, scalability, and cost-effectiveness across cloud environments.
- Skilled in architecting high-performance ETL/ELT pipelines using AWS Glue, Redshift, Snowflake, Databricks, Apache Spark, and Airflow, processing multi-terabyte structured, semi-structured, and unstructured datasets.
- Strong track record of data modeling (Star/Snowflake), query optimization, and performance tuning, reducing analytical query latency by up to 40% in enterprise systems.
- Hands-on experience operationalizing AI/ML pipelines (TensorFlow, PyTorch, Scikit-learn), enabling predictive maintenance, fraud detection, demand forecasting, and anomaly detection in production.
- Expertise in real-time data processing using Kafka, Flink, and Python, powering mission-critical applications in finance, aviation, and retail domains.
- Adept at integrating SAP ERP/Business Warehouse with modern cloud data lakes, unifying real-time and historical analytics for enterprise decision-making.
- Experienced in DevOps for data engineering — built CI/CD pipelines with Jenkins, Terraform, and Azure DevOps to automate deployments, reducing manual effort by 60%.
- Delivered impact across banking, aviation, and healthcare industries, ensuring compliance with regulatory standards (Basel, KYC/AML, FAA).
- Recognized for bridging business and technology, ensuring data solutions align with strategic objectives while enabling faster, data-driven decision-making.
- Strong collaborator in global teams across US, UK, India, and China time zones, ensuring seamless delivery of enterprise-scale projects.
- Committed to continuous learning and innovation, quickly adapting to new technologies and cloud environments, driving transformation in data architecture.
- Hands-on with ETL/ELT workflows (AWS Glue, Informatica, Airflow), real-time data processing (Kafka, Flink, IBM Streams), and BI/reporting tools.
- Strong background in data governance, audit readiness, and regulatory alignment (Basel compliance, AML/KYC standards).
- Adept at working in Agile environments, collaborating with cross-functional teams to deliver scalable and business-aligned data solutions.

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TECHNICAL SKILLS:

Big Data & Processing Frameworks	Apache Spark (PySpark, Spark SQL), Hive, Impala, Hadoop (HDFS, MapReduce), Apache Flink, IBM Streams.
Cloud Platforms	**AWS**: Glue, Redshift, S3, EC2, VPC, Lambda, RDS, CloudWatch. **Azure**: (Basic exposure) Azure DevOps. **GCP**: Big Query, DataProc, Cloud Storage, Cloud Composer, Dataflow. **Snowflake**: Warehousing, Stored Procedures, Semi-Structured Data Handling (JSON, XML, Parquet), Performance Tuning.
Programming Languages	Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch), SQL, PL/SQL, PySpark, Shell Scripting.
Data Engineering & ETL Tools	Azure Data Factory, AWS Glue, Informatica PowerCenter, SAP Data Services.
Databases	MS SQL Server, Snowflake, AWS Redshift, Azure SQL Database, MySQL, PostgreSQL.
Version Control & Workflow	Git, GitHub, Bitbucket, SVN, Agile/Scrum, Jenkins.
Data Visualization, Data Analysis & Reporting	Advanced Excel, Tableau, Power BI, KPI Tracking, Trend Analysis
Containerization & Orchestration	Docker, Kubernetes, Jenkins, Azure DevOps Pipelines, Terraform.
Specialized Skills	SAP Data Integration, Data Modeling (Star & Snowflake Schemas), Performance Tuning & Query Optimization, Data Quality & Validation, Real-time Data Processing, AI/ML Integration into Pipelines, Data Governance & Compliance (Finance, Aviation).

PROFESSIONAL EXPERIENCE

Tech Mahindra – Dallas, TX **Aug 2024 – Recent**
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Key Responsibilities and Contributions:

- Built and optimized AWS Glue data pipelines processing transactional, customer, and financial data for regulatory reporting and risk analysis.
- Automated ETL workflows across AWS Glue, Redshift, and Snowflake, ensuring timely and accurate data delivery for compliance and audit readiness.
- Designed and tuned Redshift and Snowflake stored procedures to support fraud detection, AML monitoring, and credit risk calculations.
- Parsed and transformed semi-structured XML/JSON datasets in Snowflake for structured analysis in regulatory reporting.
- Engineered large-scale AWS EMR pipelines using PySpark, Hive, and Impala to process high-volume payment, credit, and customer behavioral data.
- Created fact and dimensional models in SQL Server and Snowflake for Basel compliance, financial reporting, and customer 360 initiatives.
- Wrote complex SQL queries, joins, CTEs, aggregations, and window functions across Redshift, RDS, and Athena.
- Leveraged Informatica PowerCenter to integrate core banking, credit card, and loan systems into centralized warehouses.
- Applied data validation and quality checks to ensure accuracy and consistency of financial datasets.

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- Collaborated with risk and compliance teams to deliver analytics solutions for fraud detection, transaction monitoring, and KYC/AML reporting.
- Developed interactive Power BI dashboards with advanced features (drilldowns, bookmarks, conditional formatting) to drive actionable insights.
- Managed end-to-end reporting lifecycle and automated deployments via Jenkins CI/CD pipelines.

Environment: AWS (Glue, Redshift, EC2, S3, VPC), Snowflake, SQL Server, Matillion ETL, Informatica PowerCenter, Spark, Hive, Impala, Python, PL/SQL, Jenkins, JIRA

Inventory Locator Service (ILS) - Memphis, Tennessee **Jan 2023 – Jul 2024**
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Responsibilities:

- Developed and maintained high-throughput data pipelines using Python and PL/SQL to process aviation parts, repair orders, and marketplace transaction data.
- Designed and implemented ETL/ELT workflows using Azure Data Factory and Databricks to integrate and standardize data from multiple aviation and aerospace systems into centralized Azure Data Lake.
- Built and deployed AI/ML models leveraging Azure Machine Learning, TensorFlow, Scikit-learn, and PyTorch to enhance marketplace intelligence, fraud detection, and demand prediction for aviation components.
- Engineered big data processing solutions in Azure Databricks using Spark and Python for predictive maintenance, anomaly detection, and pricing analytics.
- Implemented data validation and monitoring scripts in Python to ensure data accuracy, reliability, and consistency across high-volume aviation transactions.
- Collaborated with Data Scientists to operationalize ML models and integrate them into production Azure pipelines for aviation aftermarket applications.
- Partnered with BI and Product teams to deliver data models, interactive dashboards, and advanced reporting via Power BI, supporting aviation part tracking and marketplace insights.
- Optimized complex PL/SQL and Spark SQL queries to improve processing performance for large aviation component and inventory datasets.
- Automated routine data processing workflows using Azure Data Factory, Databricks, and Python, reducing manual intervention and improving pipeline reliability.
- Applied data governance and compliance best practices to align with aviation and aerospace industry regulatory standards.
- Supported AI/ML-driven forecasting models deployed in Azure ML to improve aviation part availability predictions and reduce turnaround times for repair orders.
- Conducted performance tuning, query optimization, and pipeline monitoring across multiple Azure data services to enhance system efficiency.
- Collaborated with cross-functional teams to integrate AI and ML capabilities into Azure-based aviation digital platforms, improving customer decision-making and marketplace efficiency.

Environment: Python, PL/SQL, Azure Data Factory, Azure Databricks, Azure Data Lake Storage, Azure SQL Database, Azure Machine Learning, Spark, TensorFlow, Scikit-learn, PyTorch, Power BI, ETL/ELT workflows, Data Validation & Monitoring, Data Governance & Compliance Standards.

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Cognizant – Hyderabad, India **Apr 2020 – July 2022**
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Responsibilities:

- Assisted in designing and implementing ETL workflows using Apache Airflow within GCP (Google Cloud Platform) to automate data pipeline tasks.
- Developed Python scripts for data validation and transformation in Cloud Dataflow, ensuring accuracy between raw source files and Big Query tables.
- Built PySpark jobs to cleanse and merge structured/unstructured datasets, improving data readiness for analytics.
- Supported Snowflake data engineering tasks, including writing SQL queries and stored procedures for reporting and data transformations.
- Gained hands-on experience with GCP services (Big Query, Cloud DataProc, Cloud Storage, Cloud Composer) for cloud-based data integration and analytics.
- Assisted with real-time data ingestion and processing using Apache Flink and IBM Streams, contributing to event-driven analytics solutions.
- Worked with Redis for caching and Pub/Sub messaging to support real-time event-driven use cases.
- Translated Hive/SQL queries to PySpark, Data Frames for scalable big data processing.
- Performed query optimization and performance tuning in SQL/PLSQL for Oracle, Snowflake, and Big Query.
- Supported data migration initiatives from on-premises databases (Oracle, SQL Server, Postgres) to cloud data warehouses (Snowflake, Big Query).
- Collaborated with senior engineers in CI/CD pipelines (Jenkins) for automated testing and deployment of data workflows.
- Contributed to Agile ceremonies (sprint planning, daily scrums) and maintained documentation for ETL workflows and troubleshooting.

Environment: GCP (Big Query, DataProc, Cloud Storage, Cloud Composer, Dataflow), Apache Airflow, PySpark, Python, SQL/PLSQL, Snowflake, Hive, Oracle, SQL Server, Redis, Jenkins.

EDUCATION

Master’s in Information Technology Management
Webster University, Orlando, FL
Graduated: March 2024

Bachelors in Computer Science & Engineering
Aurora’s Technological & Research Institute
Graduated: July 2020